Installations

Anaconda

First of all, during this course we will use Spyder IDE from Anaconda package.

If you already have an Anaconda installed, check for updates and go to the next step.

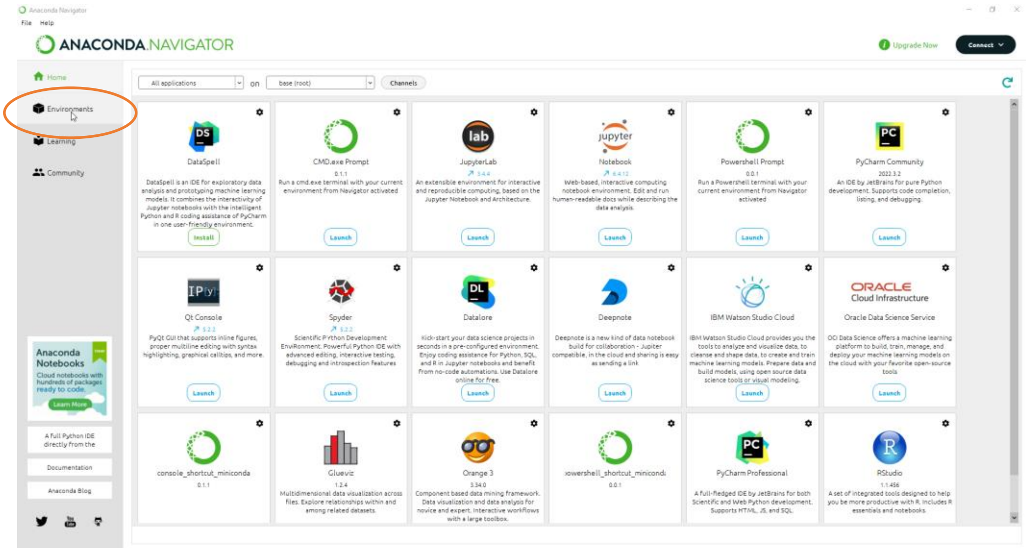
If not, then download Anaconda and install it. You can find distributive here:

https://www.anaconda.com/download

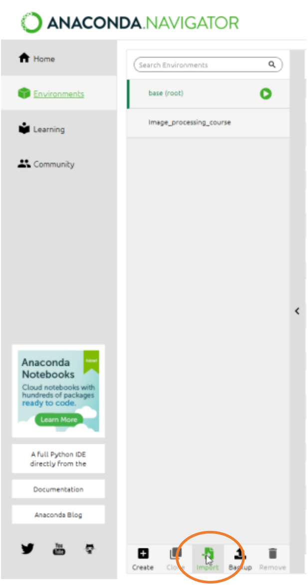
*Choose appropriate version depending on your operation system.

Environment

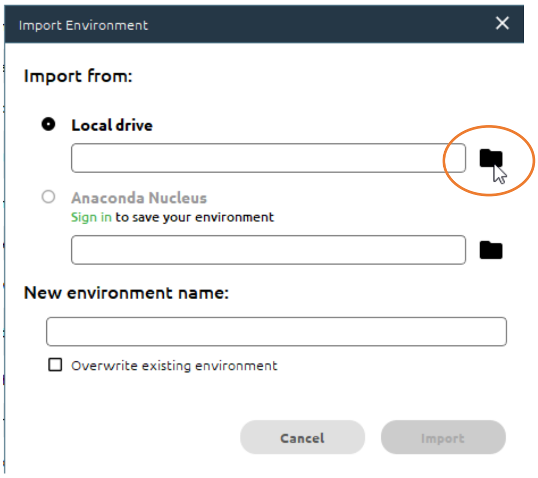
For this course we recommend to use prepared environment. To get it on your computer, download and extract archive image_processing_course to the root of D:\ drive. (You can extract it anywhere else, but then you'll need to update all paths). Then open Anaconda navigator and choose environments tab.



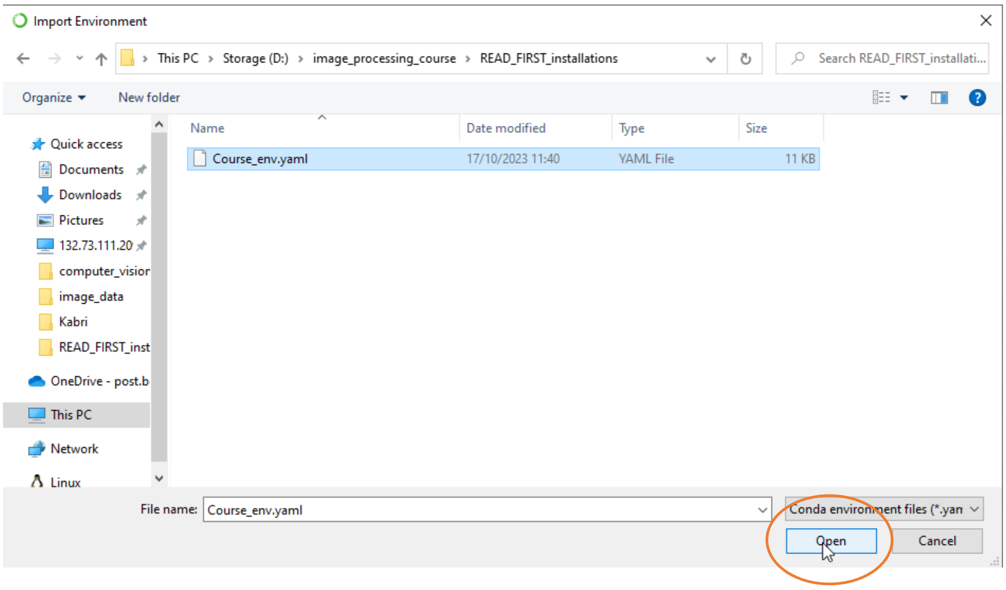
Choose Import.



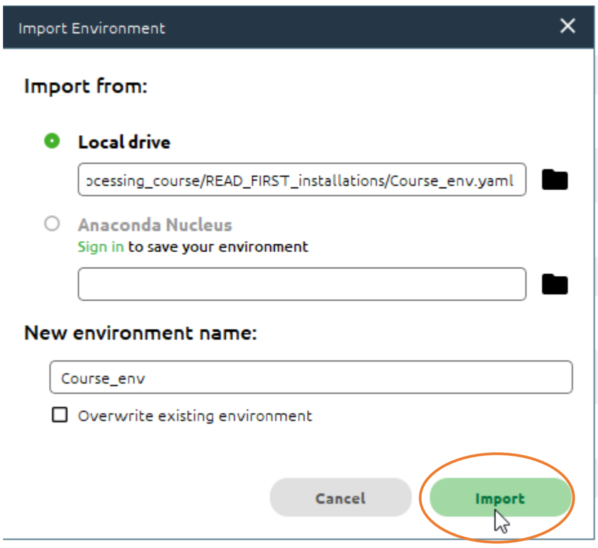
In the Import window click the folder icon next to Local drive.



Browse to D:\image_processing_course\READ_FIRST_installations and choose Course_env.yaml.



Define the name or leave it unchanged and press Import.



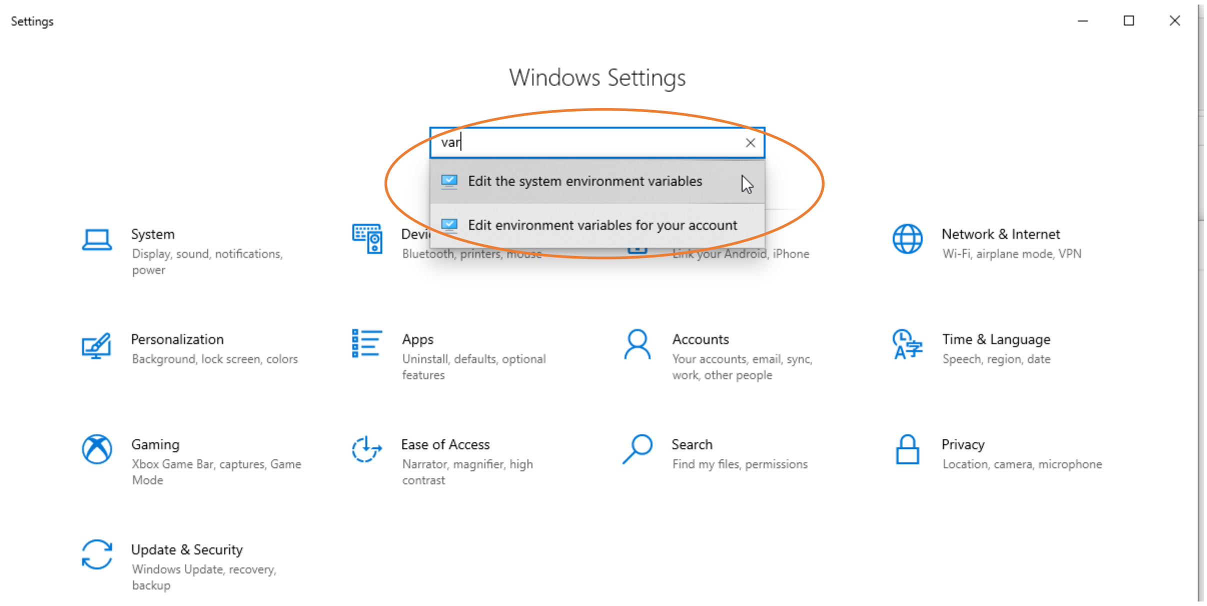
Importing will take some time (about 20 minutes on my computer).

Tensorflow

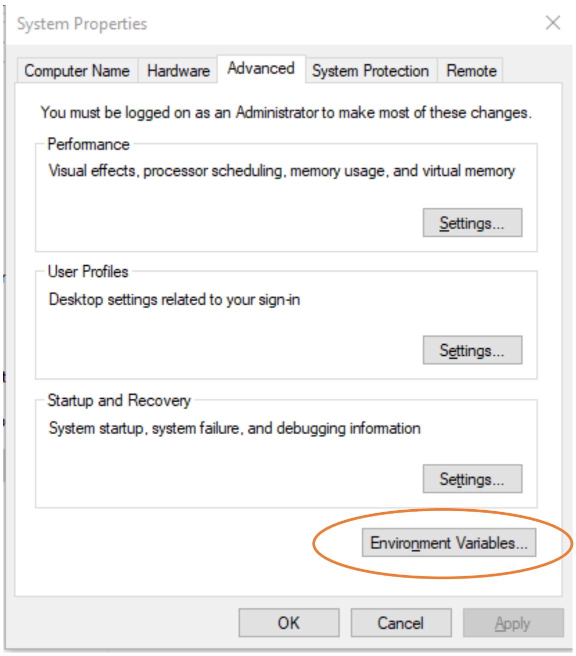
Next, we need to upload some models to tensorflow. They are already existing in archive, but we need to make them available for python.

IMPORTANT this step requires the administrator role!

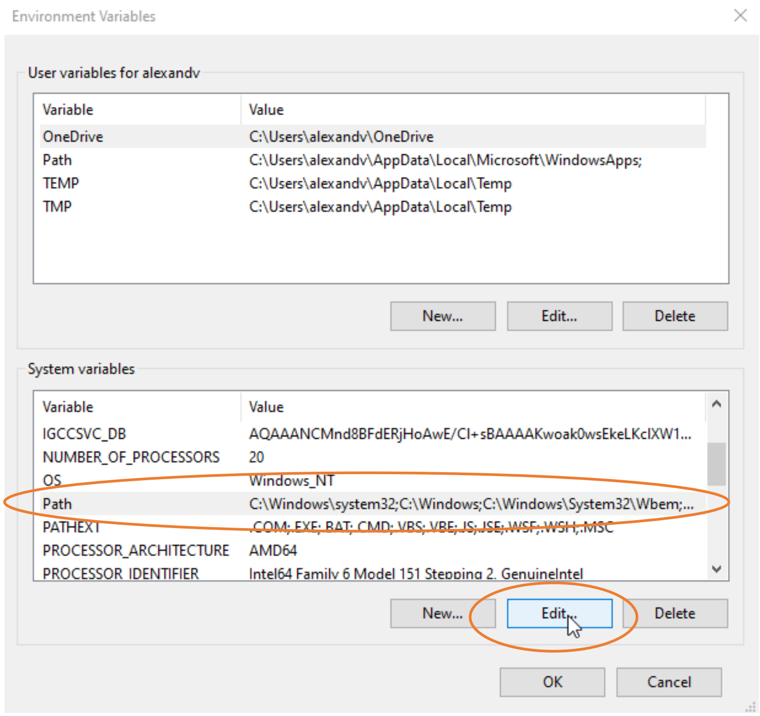
First you need to update Path variable if you are working on windows. To do it go to windows settings and find Edit the system variables.



Choose Environment Variables…

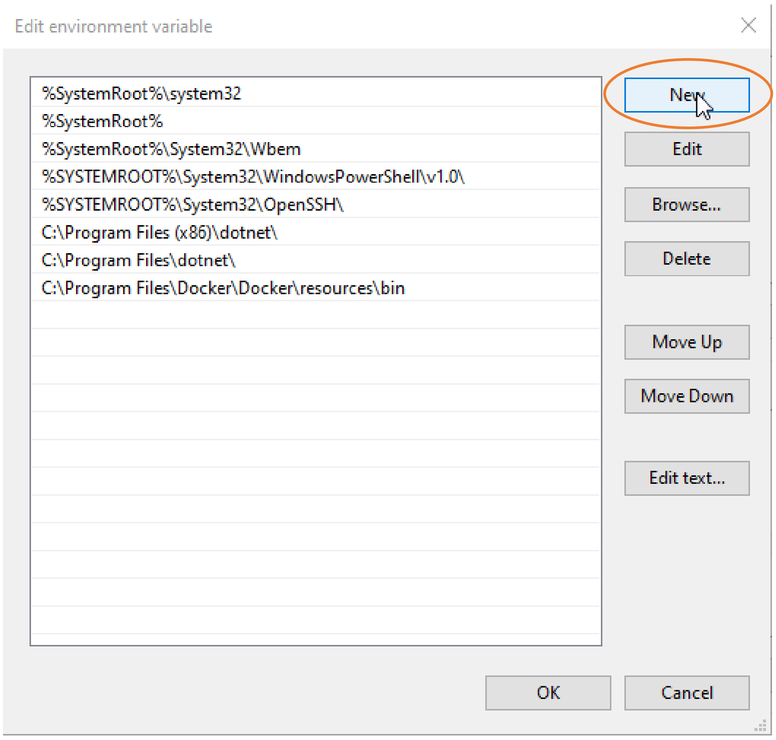


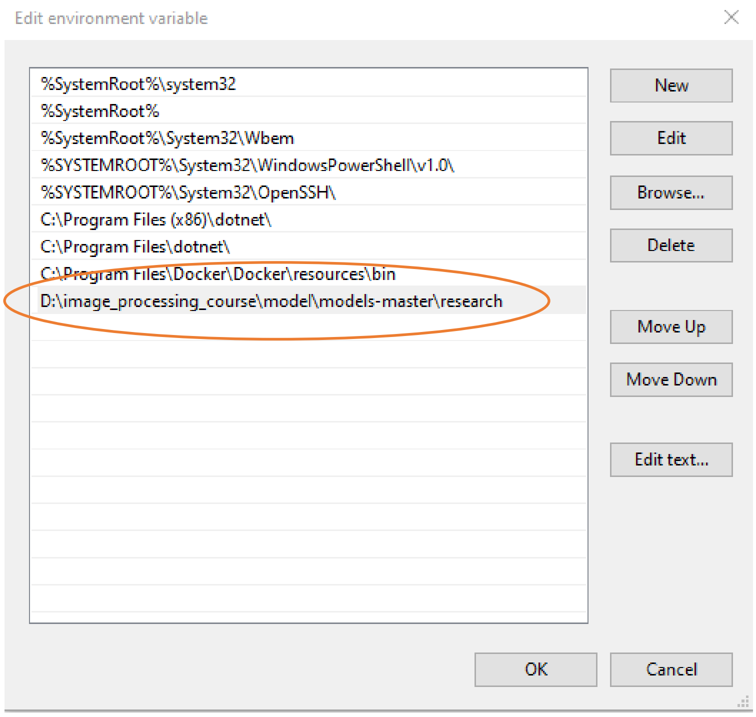
In System Variables list choose Path and click Edit…



Click New and paste following:

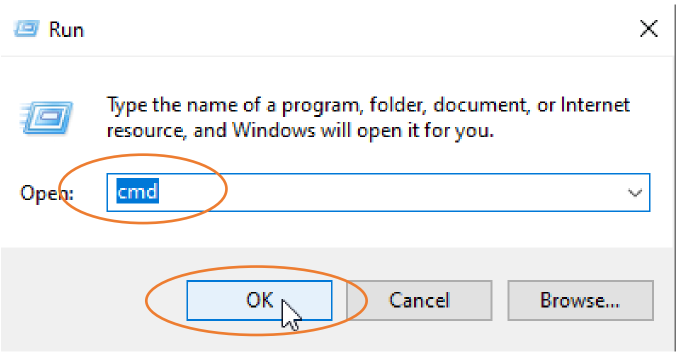
D:\image_processing_course\model\models-master\research





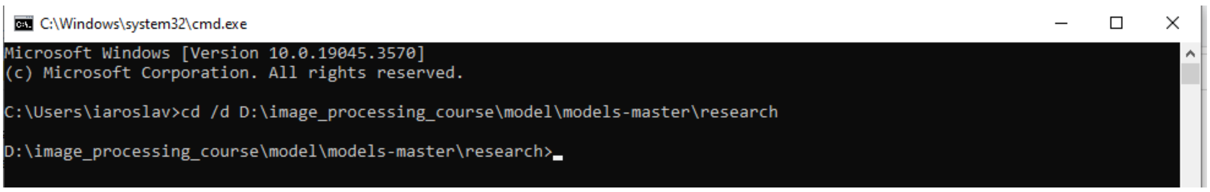
Now click Ok to close all windows.

Next, run command line (win+r, type cmd, enter).



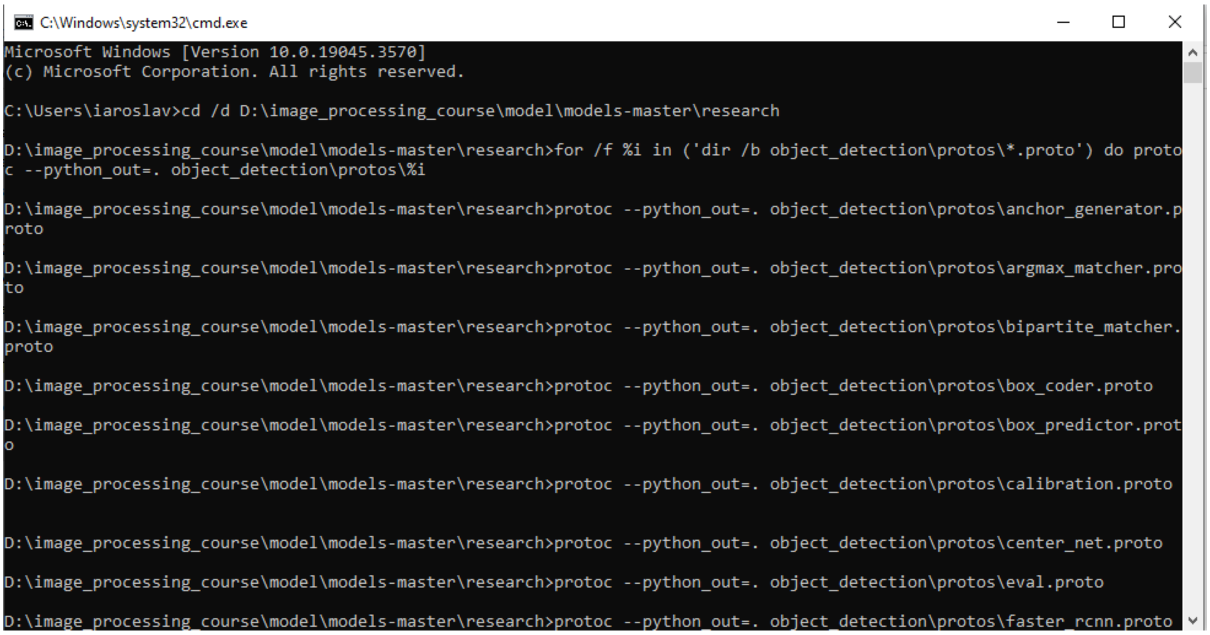
Paste and then press Enter following:

cd /d D:\image_processing_course\model\models-master\research



Paste and then press Enter following:

for /f %i in ('dir /b object_detection\protos*.proto') do protoc --python_out=. object_detection\protos\%i



After the log will finish, you are ready to work.

Don’t forget to activate your environment before running Spyder. To run Spyder, go to the Home tab of Anaconda Navigator and press launch. (For the first time you’ll probably have an Install button, press it).